

**TS 2020**

**Grade 12**

**Chemistry, Paper -II**

**Time: 3 Hours**

**Max. Marks: 60**

**Note:** Read the following instructions carefully:

1. Answer all the questions of Section — A. Answer any six questions in Section — B and any two questions in Section — C.
2. In Section — A, questions from Sr. Nos. 1 to 10 are of "Very Short Answer Type". Each question carries two marks. Every answer may be limited to two or three sentences. Answer all these questions at one place in the same order.
3. In Section — B, questions from Sr. Nos. 11 to 18 are of "Short Answer Type". Each question carries four marks. Every answer may be limited to 75 words.
4. In Section — C, questions from Sr. Nos. 19 to 21 are of "Long Answer Type". Each question carries eight marks. Every answer may be limited to 300 words.
5. Draw labelled diagrams wherever necessary for questions in Sections — B and C.

## SECTION — A

$10 \times 2 = 20$

**Note:** Answer all the questions:

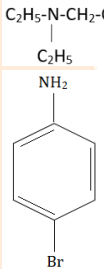
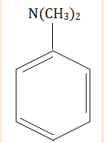
1. State Raoult's Law.
2. State Faraday's Law
3. Write ores with formulae of the following metals:
  - (i) Aluminum
  - (ii) Iron
4. What is Lanthanoid contraction?
5. What are Polymers? Give example.
6. What is vulcanization of rubber?
7. What are antacids? Give example.
8. What is tincture of iodine? What is its use?
9. What are Enantiomers?
10. What are ambident nucleophiles

## SECTION — B

$6 \times 4 = 24$

**Note:** Answer any six questions:

11. Derive Bragg's equation.
12. Define molarity. Calculate the molarity of a solution containing 5 g of NaOH in 450 ml solution.
13. What are different types of adsorption? Give any four differences between characteristics of these different types.
14. Explain the purification of sulphide ore by Froth floatation method.
15. Explain the terms:
  - (a) Ligand
  - (b) Co-ordination number

- (c) Co-ordination entity  
(d) Central metal atom/ion
16. How are  $\text{XeF}_2$  and  $\text{XeF}_4$  prepared? Give their structure.
17. What are hormones? Give one example for each.
- (a) Steroid hormone  
(b) Polypeptide hormones and  
(c) Amino acid derivatives
18. Write the IUPAC names of the following compounds:
- (a)  $\text{NH}_2\text{-CH}_2\text{-CH=CH}_2$
- (b) 
- (c) 
- (d)

## SECTION – C

$2 \times 8 = 16$

**Note:** Answer any two questions:

19. (a) What are Galvanic Cells? Explain the working of a galvanic cell with a neat sketch taking Daniel cell as example.
- (b) What are fuel cells? Give the construction of  $\text{H}_2$ ,  $\text{O}_2$  fuel cell.
20. (a) How is Ammonia manufactured by Haber's Process?
- (b) How does Ozone react with the following:

- (i) PbS
- (ii) KI
- (iii) Hg
- (iv) Ag

21. Explain the following reactions with equations:

- (i) Hell-Volhard-Zelinsky reaction (HVZ)
- (ii) Decarboxylation
- (iii) Aldol condensation
- (iv) Gatterman-Koch reaction
- (v) Ag